The Neurophysiology of Cybersickness in Virtual Reality: A Systematic Review of Neural Correlates and Neurostimulation

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***Abstract*—Cybersickness remains a major barrier to the widespread adoption of virtual reality (VR). Current subjective questionnaires are retrospective and insufficient for the dynamic, continuous assessment required to diagnose its determinants and implement real-time mitigation. To address this, the field has increasingly investigated neuromarkers: signals derived from non-invasive neuroimaging as well as neurostimulation. This PRISMA-guided systematic review synthesizes 92 studies investigating the neurophysiology of visually induced motion sickness (VIMS) in VR. The corpus explicitly unites recording modalities (77 electroencephalography, 4 functional near-infrared spectroscopy) with neurostimulation interventions (11 studies spanning galvanic vestibular, transcranial direct-current, and transcranial alternating-current stimulation) under a unified frame.**

Synthesis of the literature reveals consistent neural correlates alongside a critical methodological confound. VIMS is predominantly associated with increased low-frequency (delta and theta) power and decreased alpha-band power. However, we report meta-analytic evidence that VR hardware significantly modulates these band-power responses ($p = 0.004$ on alpha, large effect; $p = 0.015$ on beta), indicating that prior contradictions in the literature are partly attributable to hardware confounds rather than to differences in VIMS per se.

Despite these advances, the literature faces substantial methodological and reporting barriers. A methodological audit of the 38 classification or prediction studies reveals sparse reporting and arbitrary thresholds: only one study reports leave-one-subject-out validation, nineteen do not disclose a sick-versus-non-sick decision threshold, and thirteen report accuracy without sensitivity, specificity, AUC, or F1. We conclude with a call for standardized reporting, multi-site open data, and causal validation via interventional designs to establish robust neuromarkers that can enable adaptive, user-aware immersive systems.

***Index Terms*—Human-centered computing, Human computer interaction (HCI), Interaction paradigms, Virtual reality.**

I. Introduction

Virtual Reality (VR) technology has permeated numerous sectors beyond entertainment, becoming integral tools in fields such as healthcare, professional training, and education [1]. VR has been characterized as “the illusion of participation in a synthetic environment rather than external observation of such an environment,” relying on stereoscopic head-tracked displays, hand/body tracking, and binaural sound to constitute an immersive, multisensory experience [2], though no single definition has been universally adopted [3].

A barrier to widespread VR adoption is the onset of cybersickness (CS) [4]. Cybersickness is a form of visually induced motion sickness (VIMS) [5] characterized by symptoms such as nausea, dizziness, and disorientation that arise during or after immersion in a virtual environment [4]. The relationship between cybersickness, VIMS, simulator sickness, and motion sickness is treated in detail in Section II; the present review uses VIMS as the umbrella term when introducing constructs and uses the term each cited study employs when reporting its findings. Cybersickness affects a substantial portion of the user population, with studies reporting prevalence between 60% and 95% and premature session termination in 6–12.9% of participants [4]; cybersickness also degrades user experience and impairs task performance [6]. Assessment of VIMS has historically relied on subjective post-hoc questionnaires such as the Simulator Sickness Questionnaire [7]. Such instruments are retrospective, prone to recall bias, and do not capture the continuous nature of the experience [8], [9]. This limitation creates a need for objective measures grounded in the user’s physiology and for interventions that can modulate the user’s state during exposure rather than after the fact.

Two converging lines of research have responded by examining the brain. The first records neural activity, primarily electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), to identify signals that index the onset, severity, or trajectory of VIMS. The second applies non-invasive brain stimulation, including galvanic vestibular stimulation (GVS), transcranial direct-current stimulation (tDCS), and transcranial alternating-current stimulation (tACS), to modulate the underlying processes that produce sickness. We use the term *neuromarkers* to refer to signals derived from either non-invasive recording or non-invasive modulation of brain activity in this context. In a VR context, neuromarkers can carry information that subjective questionnaires cannot: continuous, time-resolved indices of user state that can in principle drive adaptive content, comfort thresholds, and closed-loop mitigation [10], [11].

In a previous review, Chang et al. [8] synthesized 33 EEG studies published before September 2021 and characterized the EEG analysis pipeline used in cybersickness research. The present systematic review differs in four respects. First, the scope extends beyond EEG to include functional near-infrared spectroscopy and non-invasive neurostimulation, consistent with the integrated frame of neuromarkers. Second, the corpus is larger and more recent: 92 studies retrieved up to 27 March 2025 from three curated databases (PubMed, Web of Science, Scopus), compared with 33 studies retrieved from Web of Science and Google Scholar up to September 2021. Third, the synthesis includes corpus-level inferential statistics on hardware and methodological confounders, in addition to the pipeline characterization shared with prior work. Fourth, this review situates the empirical findings within the broader context of the field: the theoretical frameworks invoked to explain VIMS, the terminology used across the cybersickness, VIMS, simulator sickness, and motion sickness literatures, and a critical appraisal of recurring methodological shortcomings. Where the prior scoping review concentrates on findings and pipelines, this review additionally examines the conceptual and methodological scaffolding in which those findings are produced.

The contributions of this review are threefold. First, we provide a systematic synthesis of VIMS neuromarkers in VR across 92 studies, covering neuroimaging and neurostimulation under a unified frame. Second, we report meta-analytic evidence that VR display hardware systematically modulates the neural responses associated with VIMS, recasting prior contradictions in the literature as partly attributable to hardware confounds rather than to differences in VIMS per se. Third, we conduct a methodological audit of the field's experimental designs and classification pipelines, identifying critical gaps in cross-subject validation and threshold reporting; we derive concrete reporting recommendations from this audit to guide future VR-EEG and VR-stimulation work.

The remainder of this paper is organized as follows. Section II reviews the relevant theoretical frameworks for VIMS and the principal neuroimaging and stimulation modalities. Section III describes the review methodology and search strategy. Section IV presents the corpus, the per-modality synthesis, and the corpus-level meta-analytic findings. Section V critically evaluates the methodological challenges identified in our audit and outlines a roadmap for future work. Section VI concludes.

II. Theoretical and Methodological Background

A. Terminology and Etiological Frameworks

Before surveying the theories proposed to explain cybersickness, it is useful to delimit the construct itself and to distinguish it from related forms of sickness and from perceptual phenomena that often co-occur with it.

Motion sickness is the broadest construct, encompassing adverse responses to both physical and visually specified motion [12], [13]. Within it, visually induced motion sickness (VIMS) designates sickness elicited primarily by visual motion, including cases where physical body motion is absent or inconsistent with the visual scene [14], [15]. Cybersickness is most often treated as the VR-specific instance of VIMS; simulator sickness, gaming sickness, and related labels denote overlapping exposure contexts rather than distinct symptom categories [16]–[18]. This review adopts the above hierarchy when introducing constructs, but reports each included study using the terminology its authors chose, in order to avoid imposing a uniform vocabulary on a literature that does not yet share one.

Traditionally, these constructs have been measured using subjective questionnaires. The dominant instrument is the Simulator Sickness Questionnaire (SSQ), which yields nausea, oculomotor, and disorientation subscales [7]; it is frequently substituted by the single-item Fast Motion Sickness Scale (FMS) for fast, continuous, in-task ratings [19]. Broader diagnostic criteria for motion sickness and VIMS extend these subscales to gastrointestinal, thermoregulatory, arousal, dizziness or vertigo, headache, and visual or oculomotor domains [15].

Several etiological frameworks have been proposed to explain cybersickness and the wider VIMS phenomenon; they are not mutually exclusive and often emphasise complementary mechanisms.

The most commonly used framework is the *sensory conflict theory*, which attributes sickness to contradictory visual, vestibular, and proprioceptive signals, or between current sensory input and the prior exposure history encoded by the observer [20]–[22]. Two refinements within this family further specify the nature of the conflict: subjective vertical mismatch frames it as a discrepancy between sensed and expected gravitational vertical [23], while rest-frame conflict locates it in the observer's difficulty identifying a stable spatial reference [24]. This family is particularly pertinent to VR, where HMDs routinely present visual self-motion while vestibular and proprioceptive signals indicate little or no corresponding physical movement.

A second framework, *postural instability theory*, locates the cause of sickness not in sensory mismatch but in a sustained inability to maintain postural control under unfamiliar sensorimotor demands [25], [26]. The theory is most directly testable in VR paradigms involving standing, walking, or large-field optic flow, where body sway and control dynamics can be measured objectively and have been reported to precede or accompany subjective sickness [27]. A related strand of recent VR work returns to the role of vection itself, showing that sickness intensifies when visually induced self-motion violates the user's expected motion or control state [28], [29].

A further set of proposals address narrower mechanisms or different levels of explanation. Vergence-accommodation conflict identifies a mismatch between binocular convergence and the fixed focal distance of stereoscopic displays as a source of oculomotor discomfort [30], while oculomotor theories more broadly relate sickness to specific eye-movement patterns, such as nystagmus or ocular torsion, evoked by the visual stimulus [31]. At an evolutionary level, Treisman's toxin-defense hypothesis explains why disturbed sensory integration should give rise to nausea and emesis, but does not specify the proximate sensory conditions that trigger them [32]. Finally, predictive-coding formulations recast sensory mismatch as a persistent prediction error between expected and observed input, offering a unifying cognitive framework for the preceding theories [33].

B. Neural Recording and Modulation Modalities

VIMS has historically been assessed through subjective reports, but a growing body of work uses neural recordings, and more recently non-invasive brain stimulation, to characterise the user's state during VR exposure or to actively modulate it.

Electroencephalography (EEG) is the most widely used recording modality in this literature. Its high temporal resolution supports the study of spectral power, event-related potentials, and connectivity dynamics throughout the development of sickness, and its portability makes it compatible with HMD-based and other interactive VR paradigms [27], [34]. These strengths are offset by noise, limited spatial specificity, particularly for deep vestibular structures, and by methodological sensitivities, such as motion artifacts, interference, reference choice, and baseline selection, which constrain the interpretation of reported effects.

Functional near-infrared spectroscopy (fNIRS) is an emerging optical technique that measures cortical hemodynamic changes via near-infrared light [35], [36]. It offers an intermediate trade-off between temporal and spatial resolution and tolerates moderate movement, making it well suited to VR paradigms such as HMD-based roller-coaster exposure, VR locomotion, or stereoscopic viewing [11], [37]–[39]. Because it relies on hemodynamic rather than electrical signals, its temporal resolution is substantially lower than that of EEG, and its optical penetration restricts measurement to superficial cortical regions.

Beyond recording, non-invasive neurostimulation has begun to be used to modulate sickness-relevant processes during VR or visually induced motion exposure. Reported approaches include galvanic vestibular stimulation, transcranial direct-current stimulation, transcranial alternating-current stimulation, and related techniques [40]. By directly perturbing vestibular or cortical activity, these methods provide a causal complement to the correlational evidence offered by EEG and fNIRS.

III. Review Methodology

A. Search Strategy

A systematic literature search was conducted across three databases: PubMed, Web of Science, and Scopus. A publication-date cutoff of 27 March 2025 was applied to PubMed and Web of Science; no earlier date restriction was imposed, so the search captures the full historical record of the field. The query targeted the intersection of three concept blocks: brain-directed measurement or modulation methods, cybersickness and related sickness constructs, and immersive virtual reality. The full query was:

```
(("Brain Computer Interface" OR BCI OR EEG OR
    electroencephalogram OR ERP OR MEG OR
    magnetoencephalography OR fMRI OR ECoG OR
    fNIRS OR PET OR "positron emission
    tomography" OR neurostimulation OR "brain
    stimulation" OR "non-invasive brain
    stimulation" OR tACS OR "transcranial
    alternating current stimulation" OR tDCS
    OR "transcranial direct current
    stimulation" OR TMS OR "transcranial
    magnetic stimulation" OR "galvanic
    vestibular stimulation" OR GVS)
AND (Cybersickness OR "simulator sickness" OR
    "visually induced motion sickness" OR VIMS
     OR "motion sickness")
AND ("virtual reality" OR VR OR "head-mounted
    display" OR HMD OR "immersive environment
    "))
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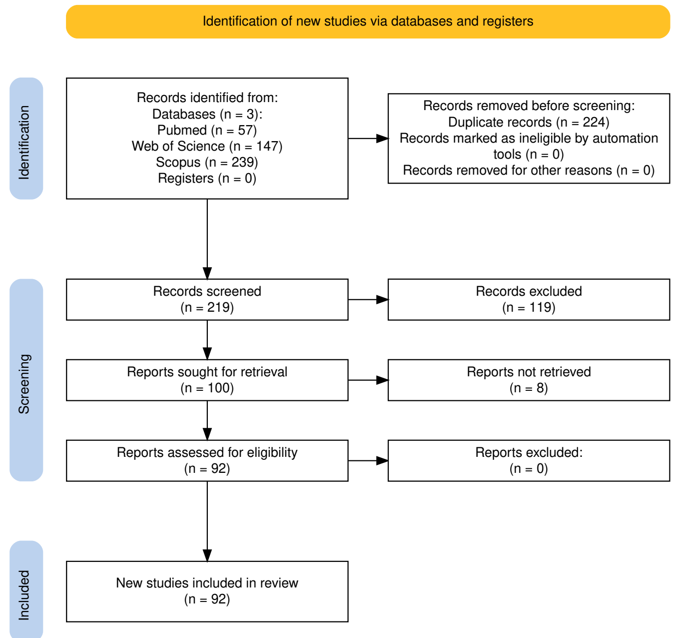

Fig. 1: PRISMA flow diagram for the present review.

B. Study Selection and Eligibility Criteria

Records identified by the search were screened in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement; the full selection process is summarised in the PRISMA flow diagram (Figure 1). The combined search returned 443 records: 57 from PubMed, 147 from Web of Science, and 239 from Scopus. Duplicate removal across the three sources reduced this set to 219 unique records.

Each of the 219 unique records underwent manual title-and-abstract screening against the following eligibility criteria. A record was included if it met all of the following:

- The study employed a neuroimaging modality (EEG, ERP, fNIRS, fMRI, MEG, ECoG, PET) or a brain-stimulation modality (tDCS, tACS, TMS, GVS) to measure or modulate brain activity.
- The study addressed VIMS or a closely related construct (simulator sickness, or motion sickness in a virtual environment) within a virtual reality or otherwise immersive context.
- The study reported original empirical data from human participants.

A record was excluded if it met any of the following:

- The work was non-empirical (literature reviews, meta-analyses, editorials, letters, or conference abstracts without an accompanying full paper).
- The study did not address VIMS or a related sickness construct as a primary outcome.
- The full text could not be retrieved.

Title-and-abstract screening excluded 119 records, leaving 100 reports sought for full-text retrieval. Of these, 8 could not be obtained as full text and were excluded at the retrieval stage. The remaining **92 articles** comprise the final corpus analysed in this review.

C. Modalities and Dataset-Reuse Studies

Two aspects of the corpus warrant explicit comment before the synthesis: the modalities that the query covered but that returned no eligible studies, and the inclusion of studies that rely on previously released datasets.

The query included PET, MEG, and fMRI terms alongside the more represented modalities. PET returned no eligible studies; MEG returned a single record, which was excluded at the title-and-abstract stage as not addressing VIMS or a related sickness construct; fMRI returned no records at all. The absence of these modalities from the final corpus therefore reflects the current literature rather than a scope restriction imposed by the search.

A subset of retrieved studies train machine-learning classifiers on previously released datasets without recording new data. Pure-dataset benchmarking is a related but methodologically distinct literature, since its primary objective is classification performance rather than the identification or modulation of neuromarkers. Six papers in the final corpus reuse one or more publicly released datasets (in two cases combined with newly recorded data); they were retained because they satisfied all eligibility criteria and are flagged where relevant in the synthesis.

D. Data Extraction and Thematic Synthesis

For each of the 92 included articles, a structured data extraction process was performed. This process was guided by a predefined protocol designed to ensure consistency and to facilitate subsequent comparison and synthesis. The following data points were extracted from each paper:

1) **Summary of Contribution:** A concise summary of the study's primary objective and its main contribution.
2) **Sickness Measurement:** The methods used to measure or label VIMS, distinguishing between subjective questionnaires (e.g., SSQ, FMS, VRSQ) and objective physiological or behavioural metrics (e.g., galvanic skin response, postural sway).
3) **Methodological Pipeline:** A breakdown of the neurophysiological pipeline, including: (a) the measurement or stimulation modality, (b) the key feature extraction techniques (e.g., power spectral density, event-related potentials, functional connectivity), and (c) the primary analysis or classification methods (e.g., statistical tests, machine-learning models).
4) **Theoretical Framework:** The explicitly stated or implicitly assumed theory of VIMS (e.g., sensory conflict, postural instability) that guided the study's design or interpretation.
5) **Quantitative Performance:** Reported performance measures, such as classification accuracy, correlation coefficients, root-mean-square error, or statistical significance values.
6) **Empirical Study Details:** For empirical studies, specific details of the experimental protocol, including the VR environment, display hardware, sickness-inducing stimulus, and number of participants.
7) **Reported Shortcomings:** Limitations, caveats, or weaknesses identified by the authors or apparent from the study's design.

Following extraction, each paper was assigned to a primary research-objective category to support quantitative synthesis. An initial set of categories was defined *a priori* based on established research paradigms and refined iteratively during extraction. The final categories, which structure the Results section, are: Classification (Binary, Likert, Continuous), Correlates (Brain Regions, Rhythms), Mitigation (A/B Testing, Real-time, Neurostimulation), User Differences, Dataset Provision, Recovery, and studies on the Effects of VIMS on other cognitive or performance metrics.

IV. Results

The 92 corpus studies address four main research objectives: classification or prediction of VIMS from physiological signals (38 studies), identification of neural and physiological correlates (25), development of mitigation strategies (17), and characterisation of individual susceptibility (3). Nine residual studies (cognitive effects, recovery, dataset contributions, other framings) are integrated into the relevant subsections, with full discussion deferred to section V. This section adds two cross-cutting layers on top of these objectives: subsection IV-A describes the corpus as a whole, and subsection IV-F reports meta-analytic observations across the 92 studies.

Activity in the field has accelerated in recent years: 63.0% of the corpus (58 of 92 studies) was published from 2021 onwards, with a corresponding shift toward classification, mitigation, and susceptibility objectives (Figure 2). Per-study sample sizes have also increased over the same period (Mann–Whitney U on studies published before 2021 versus from 2021 onwards, Cliff's $\delta = -0.337$, $p = 0.007$, medium effect).

Table I lists all 92 studies with their modality, sample size, virtual reality hardware, stimulus paradigm, VIMS measure, primary objective, and topic.

TABLE I: All 92 studies included in this systematic review. Modality: EEG, fNIRS, fMRI = neuroimaging; GVS = galvanic vestibular stimulation; tDCS/tACS = transcranial direct/alternating current stimulation; ECG, GSR, EOG, EMG, EGG = peripheral physiology; posturography = balance / centre-of-pressure recording. CS measure abbreviations: SSQ = Simulator Sickness Questionnaire; VRSQ = VR Sickness Questionnaire; FMS = Fast Motion-Sickness Scale; FMS+SSQ = repeated FMS sampling plus SSQ assessment; MSSQ = Motion Sickness Susceptibility Questionnaire; cont. = continuous self-report; Likert = single-item Likert; custom = study-specific scale; behav. = behavioural measure only; none = no cybersickness measure reported; studies listed as cont., Likert, or behav. without a named questionnaire did not report a specific instrument. Sample size N gives subjects analysed. Topic gives a short noun phrase describing the focus of each paper.

Ref.	Modality	N	VR device	Stimulus	CS measure	Objective	Topic
[41]	EEG	2	HMD	Mixed	custom	Mitigation	EEG-informed scene-simplification mitigation
[42]	EEG+EOG	17	HMD	Driving (passive)	SSQ	Correlates	Alpha connectivity and P3 in susceptibility
[43]	EEG+EOG	33	Screen	Passive viewing	FMS	Correlates	EEG time-frequency correlates of VIMS severity
[44]	EEG	31	HMD	Mixed	SSQ	Correlates	Spectral EEG correlates of VIMS in HMD
[45]	EEG+ECG+GSR+EOG	27	HMD	Active navigation	custom	Classification	Spiking-network EEG+HRV cybersickness prediction
[40]	tACS+GSR	37	HMD	Passive viewing	MSSQ	Mitigation	10 Hz tACS over vestibular cortex
[46]	EEG	68	HMD	Mixed	SSQ	Other	Sex differences in EEG neurofeedback sessions
[47]	EEG+tDCS+ECG+EOG	41	HMD	Passive viewing	SSQ	Mitigation	Placebo tDCS in EEG neurofeedback for cybersickness
[48]	EEG+ECG	39	HMD	Active navigation	SSQ	Cognition	VR-parameter manipulation during EEG neurofeedback
[49]	EEG	15	HMD	Passive viewing	SSQ	Classification	Resting-state EEG entropy classification
[50]	EEG	12	Screen	Passive viewing	SSQ	Mitigation	Rest-frame mitigation with EEG correlates
[51]	EEG+ECG+EOG	20	HMD	Passive viewing	SSQ	Correlates	Heartbeat-evoked potentials as cybersickness correlates
[52]	EEG	19	Projector	Driving (passive)	cont.	Correlates	ICA-based motion sickness EEG component clusters
[53]	EEG	19	Projector	Driving (passive)	custom	Correlates	Continuous motion sickness EEG dynamics
[54]	EEG	20	HMD	Other	SSQ	Correlates	blur-induced visual discomfort in HMDs
[55]	EEG	16	HMD	Active navigation	SSQ	Correlates	EEG during VR sickness while walking
[56]	EEG	32	HMD	Mixed	SSQ	Dataset	EEG cybersickness dataset and benchmark
[57]	EEG	32	HMD	Active navigation	SSQ	Classification	EEG-based comparison of VR locomotion techniques
[9]	EEG+ECG+EOG+EGG+ posturography	20	HMD	Active navigation	Likert	Classification	Multimodal EEG+physiology severity classification
[58]	EEG+ECG+GSR	25	Other	Passive viewing	SSQ	Classification	Convolutional-transformer EEG cybersickness detection
[59]	EEG	23	HMD	Passive viewing	SSQ	Classification	Multi-domain EEG feature comparison
[60]	EMG	24	HMD	Passive viewing	MSSQ	Correlates	VEMP changes after VR vection exposure
[61]	GVS	47	HMD	Passive viewing	SSQ	Mitigation	Omnidirectional GVS for 360° video viewing
[62]	EEG+EOG	23	HMD	Passive viewing	SSQ	Classification	LSTM regression of continuous VRMS severity
[63]	EEG	15	HMD	Mixed	SSQ	Classification	Multi-scale EEG brain-network classification
[64]	EEG	15	HMD	Passive viewing	SSQ	Classification	Bilateral asymmetry features for VRMS detection
[65]	EEG+EOG	22	HMD	Passive viewing	SSQ	Classification	Siam-LSTM with EEG functional brain networks
[66]	EEG+posturography	22	HMD	Active navigation	VRSQ	Classification	Dual-pathway raw EEG and network model
[67]	EEG	40	Screen	Passive viewing	SSQ	Indiv. diff.	EEG band-power correlates of cybersickness susceptibility
[68]	EEG	24	HMD	Passive viewing	cont.	Classification	Deep-learning EEG classification of cybersickness
[69]	EEG	9	Screen	Active navigation	SSQ	Other	Gamma-band EEG visualisation of sickness onset
[70]	EEG+ECG+GSR+EOG	45	CAVE	Active navigation	SSQ	Correlates	EEG and autonomic correlates during navigation
[71]	EEG+ECG+GSR+EOG+ EGG	57	Projector	Active navigation	SSQ	Correlates	Multimodal physiology of navigation cybersickness
[72]	EEG	202	HMD	Passive viewing	Likert	Classification	EEG-to-video transfer for cybersickness prediction
[73]	EEG+EOG	30	HMD	Passive viewing	SSQ	Correlates	Source-localised alpha cybersickness correlates after VR
[74]	EEG	16	Projector	Driving (passive)	cont.	Classification	ICA-EEG regression of motion-sickness level
[75]	EEG	6	CAVE	Driving (passive)	cont.	Classification	Genetic feature selection for EEG classification
[34]	EEG	43	HMD	Passive viewing	SSQ	Correlates	EEG band-power correlates with continuous joystick cybersickness
[76]	EEG+ECG+GSR	40	HMD; Screen	Passive viewing	SSQ	Classification	EEG and physiology fusion for SSQ estimation
[77]	EEG+EOG+posturography	20	Projector	Driving (passive)	custom	Classification	Multimodal ensemble for VIMS severity
[78]	EEG	18	HMD	Passive viewing	behav.	Classification	Wavelet-packet EEG features for VRMS
[79]	tDCS	20	HMD	Passive viewing	cont.	Mitigation	Cathodal parietal tDCS for cybersickness
[80]	EEG+EOG	22	HMD	Passive viewing	FMS	Classification	EEG vs. autonomic features for VR sickness

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Ref.	Modality	N	VR device	Stimulus	CS measure	Objective	Topic
[81]	EEG+EOG	20	HMD	Passive viewing	FMS	Indiv. diff.	EEG biomarker of VRMS resistance
[82]	EEG	39	HMD	Mixed	FMS	Classification	Resting EEG prediction of in-car VRMS
[83]	EEG+tACS	18	HMD	Mixed	FMS	Mitigation	Online 18 Hz tACS over P3
[84]	EEG	130	HMD	Passive viewing	SSQ	Classification	LSTM prediction of impending motion sickness
[85]	EEG	16	TV	Passive viewing	SSQ	Other	Test-retest reliability of EEG sickness markers
[86]	EEG	9	Projector	Driving (passive)	custom	Correlates	EEG dynamics during VR motion sickness
[87]	EEG	6	CAVE	Driving (passive)	cont.	Classification	BCI estimation of motion-sickness level
[88]	EEG+ECG	25	HMD	Passive viewing	cont.	Classification	Real-time EEG+ECG cybersickness warning
[89]	EEG	9	CAVE	Driving (active)	cont.	Correlates	Wearable EEG and cardiovascular correlates
[90]	EEG	8	Screen	Driving (active)	SSQ	Correlates	Low-channel EEG markers for VIMS
[91]	EEG	36	HMD	Passive viewing	SSQ	Classification	CNN-ECA-LSTM continuous cybersickness regression
[92]	EEG	36	HMD	Passive viewing	SSQ	Classification	EEG spectral and network regression of severity
[93]	EEG	25	HMD	Passive viewing	SSQ	Classification	Wavelet-packet GRU for VR motion sickness
[94]	EEG	9	Screen	Active navigation	SSQ	Classification	Time-domain EEG features for cybersickness
[6]	EEG+EOG	29	HMD	Passive viewing	SSQ	Cognition	P3b ERP attention markers during VR sickness
[95]	EEG+ECG+EOG	8	HMD	Active navigation	custom	Classification	Multimodal symptom-wise cybersickness markers
[96]	EEG	18	HMD	Passive viewing	cont.	Correlates	EEG correlates of display-parameter cybersickness
[97]	EEG	40	Screen	Passive viewing	SSQ	Correlates	EEG microstates and functional connectivity
[98]	ECG+GSR	13	HMD	Mixed	VRSQ	Classification	Symbolic ML on peripheral biosignals
[33]	EEG	14	HMD	Passive viewing	SSQ	Correlates	VIMS effective connectivity and predictive coding
[99]	EEG	33	HMD	Passive viewing	SSQ	Correlates	EEG across speed, complexity, and stereoscopy
[100]	EEG	9	Screen	Active navigation	SSQ	Classification	Rule-based EEG severity classification
[101]	GVS+EGG	19	HMD	Active navigation	SSQ	Mitigation	Simulator-sickness GVS mitigation during flight VR
[102]	GVS+posturography	140	HMD	Passive viewing	custom	Mitigation	GVS probing of vestibular contribution
[103]	EEG+EMG	28	HMD	Passive viewing	MSSQ	Classification	BioVRSea EEG+EMG motion sickness prediction
[104]	EEG+EOG	20	HMD	Mixed	FMS	Classification	EEG preprocessing trade-offs for cybersickness
[105]	EEG+GSR	22	HMD	Passive viewing	SSQ	Classification	Explainable AI on multimodal physiology
[106]	EEG	25	HMD	Passive viewing	SSQ	Classification	PLV connectivity CNN-LSTM for VIMS
[107]	GVS+GSR	20	HMD	Passive viewing	SSQ	Mitigation	Wearable motion-synchronised GVS
[108]	EEG	22	HMD	Passive viewing	SSQ	Correlates	EEG comparison of HMD versus screen VR
[109]	EEG	31	HMD	Passive viewing	SSQ	Correlates	EEG across VR motion parameter variations
[110]	tDCS+ECG+posturography	20	HMD	Passive viewing	SSQ	Mitigation	Anodal tDCS over right TPJ
[39]	fNIRS+ECG+posturography	11	HMD	Passive viewing	behav.	Correlates	fNIRS and posturography for 3D sickness
[111]	EEG+ECG+EGG	28	HMD	Mixed	FMS+SSQ	Indiv. diff.	EEG aperiodic activity for susceptibility profiling
[10]	EEG	53	HMD	Passive viewing	SSQ	Mitigation	Real-time EEG-driven factor-specific mitigation
[112]	EEG	6	HMD	Active navigation	SSQ	Mitigation	Gaze-contingent peripheral blur mitigation
[113]	Other	108	Projector	Mixed	SSQ	Mitigation	Bone-conducted vibration vestibular stimulation
[114]	GVS	40	HMD	Active navigation	FMS+SSQ	Mitigation	Noisy GVS for nauseogenic VR
[115]	EEG	6	Projector	Driving (passive)	cont.	Classification	Continuous EEG-based motion-sickness estimation
[116]	EEG	40	TV	Passive viewing	SSQ	Recovery	EEG recovery dynamics after VR exposure
[117]	EEG	17	HMD	Active navigation	SSQ	Cognition	Inhibition-related ERPs after VIMS
[118]	EEG	12	HMD	Other	SSQ	Cognition	Full-body exergame, simulator sickness, EEG theta
[119]	EEG	16	HMD	Passive viewing	SSQ	Classification	SE-CNN regression of VR motion sickness from EEG
[38]	fNIRS	1	HMD	Active navigation	SSQ	Mitigation	fNIRS-triggered FOV-narrowing mitigation
[11]	fNIRS	10	HMD	Mixed	SSQ	Mitigation	Wearable fNIRS for adaptive FOV restriction
[120]	EEG+ECG	64	HMD	Passive viewing	SSQ	Classification	Interpretable spiking-network EEG prediction
[121]	EEG	25	HMD	Mixed	FMS+SSQ	Mitigation	Multisensory VIMS mitigation with EEG correlates
[37]	fNIRS	30	HMD	Passive viewing	SSQ	Correlates	fNIRS angular-gyrus correlates of cybersickness
[122]	EEG+EOG	15	HMD	Passive viewing	none	Classification	S-transform EEG features for cybersickness

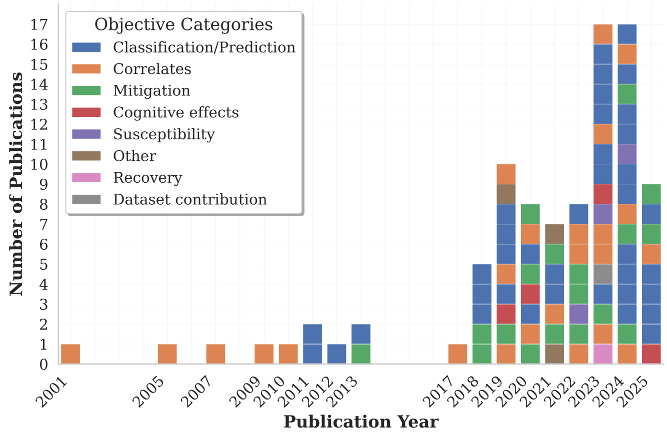


Fig. 2: Distribution of reviewed papers by year and primary research objective.

A. *Corpus Characteristics*

The 92 included studies cluster on EEG-based investigations of young adults in head-mounted VR, but vary widely on baseline condition, montage, and preprocessing. The dispersion documented in this subsection is the precondition for the meta-analytic observations reported in subsection IV-F.

1) Sample Sizes and Demographics: The 92 studies span 2001–2025. Across the corpus, 2687 participants were analysed (recruited: 2822), with a mean of 29.2 ± 29.0 participants analysed per study (median: 22, range: 1–202). The 62 studies that report an age range yield a mean age of 23.7 ± 2.7 years, indicating a focus on young-adult populations; 14 studies do not report age or report ranges that cannot be parsed.Figure 3 shows the age distribution across studies.

2) Neuroimaging and Physiological Recording Modalities: EEG is the predominant modality (77 of 92 studies, 83.7%). The remaining studies use fNIRS (4 of 92, 4.3%) or non-invasive neurostimulation without concurrent recording (11 of 92, 12.0%: GVS in 5, tDCS in 3, tACS in 2). No corpus study uses fMRI, MEG, or PET; section III states the search-query and screening implications of this absence.

EEG channel counts vary widely (median 30, range 1–256, mean 39.6 ± 55.8, $n = 72$ studies reporting). The most frequent montages are 32-channel (18 studies), 64-channel (9), 14-channel (8), 16-channel (5), and 30- or 256-channel (4 each); a long tail of low-channel configurations (4 to 16 channels) is consistent with portable in-VR recording. Figure 4 shows the distribution of EEG channel configurations across studies. Channel count is positively correlated with reported within-subject classification accuracy ($r = 0.403$, $p = 0.037$, $n = 27$); this observation is revisited in subsection IV-F.

Concurrent physiological recording is reported in 36 of 92 studies (39.1%). Most commonly used are electro-oculography (18 studies), electrocardiography (16), and galvanic skin response (9). Present but less common are electromyography (2), electrogastrography (4), photoplethysmography (7), respiration (2), skin temperature (4), and blood pressure (1).

3) Experimental Design and Data Collection Protocols: Study designs predominantly employ within-subject comparisons (78 of 92, 84.8%), with 7.6% employing between-subject designs exclusively and 7.6% using both. The majority of protocols are single-session (mean 1.6 ± 2.1 sessions, median 1, $n = 91$). Session duration averages 36.3 ± 32.6 minutes (median 25.8, range 1.8–180, $n = 76$), and the VIMS-inducing stimulus block within a session averages 15.2 ± 15.4 minutes (median 10.0, range 0.5–90, $n = 86$).

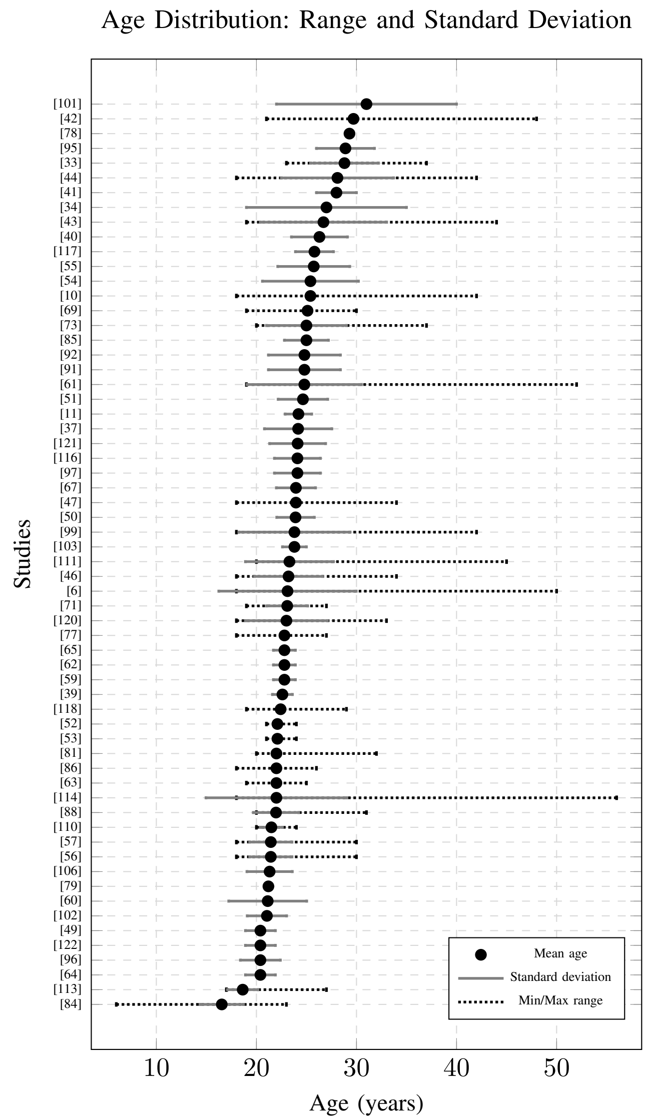


Fig. 3: Age distribution across studies showing mean age (black dots) with age ranges (dotted bars) and standard deviations when available (solid bars). The age range shows the minimum and maximum participant ages, while the standard deviation shows the typical spread around the mean age.

Virtual reality is delivered through head-mounted displays in 73.9% of studies (68 of 92), with screens (8.7%) and projection systems (8.7%) as the next most common configurations, followed by CAVE (4.3%), television (2.2%), and other or mixed-modality setups (2.2% combined). Virtual environments cluster on navigation (17 studies, any-mention 18.5%), roller-coaster experiences (16, 17.4%), games (15, 16.3%), driving (13, 14.1%), passive video (10, 10.9%), boating or flight (7,

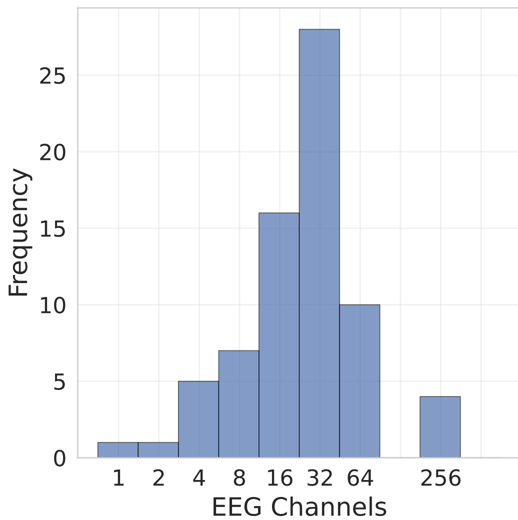


Fig. 4: Distribution of EEG channel configurations across VIMS studies.

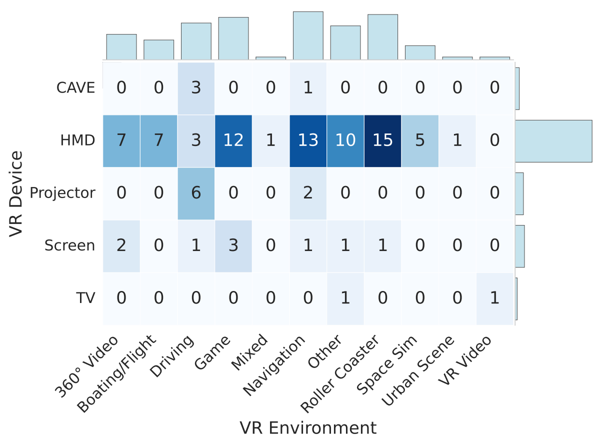


Fig. 5: Heatmap of the distribution of VR device types and virtual environments across studies. Studies employing multiple devices or environments contribute towards each.

7.6%), and space simulators (5, 5.4%). The joint distribution of device type and environment is shown in Figure 5; the hardware mix bears on the meta-analytic findings in subsection IV-F.

4) VIMS Assessment Methods: VIMS is measured through questionnaire instruments in 70 of 92 studies (76.1%), continuous ratings during exposure in 20 (21.7%), single-item Likert scales in 15 (16.3%), and behavioural indices in 8 (8.7%); these categories sum above 100% because 19 studies use more than one method on the same exposure. Among studies using questionnaires, the Simulator Sickness Questionnaire is the most common instrument (58 of 92), followed by the Fast Motion Sickness Scale (6), the Motion Sickness Susceptibility Questionnaire (3), the Virtual Reality Sickness Questionnaire (2), and other instruments (6); 17 studies do not specify the questionnaire used.

5) Baseline Conditions and Preprocessing Heterogeneity: Baseline conditions vary across the corpus: 39 of 92 studies use a task baseline, 16 use an eyes-open resting baseline, 8 use a pre-stimulus segment of the same recording, 5 use an eyes-closed resting baseline, and the remainder either do not specify a baseline (3) or do not report one (21 missing, including the neurostimulation-only studies). The five eyes-closed cases are a minority of the corpus, but alpha-band reactivity to eye closure overlaps with the alpha modulations attributed to VIMS; section V develops this critique and discusses recommended alternatives.

Across the 77 EEG corpus studies, band-pass and notch-filter reporting is heterogeneous: 11 (14.3%) do not report a band-pass filter, and 59 (76.6%) do not state a notch-filter frequency. Of the 66 that report a band-pass setting, 59 give numeric low and high cutoffs, summarised in Figure 6; among these, low cutoffs span 0–8 Hz and high cutoffs 15–100 Hz, and no configuration predominates in the corpus.

B. Neural and Physiological Correlates of VIMS

Across the 77 EEG, 4 fNIRS, and a subset of corpus studies using event-related potential analyses, the corpus converges on a low-frequency power increase during VIMS; higher-frequency and lateralised findings are less consistent at the corpus level and, as shown in subsection IV-F, are partly attributable to differences in VR hardware platform.

1) EEG Spectral Power: Across the studies reporting per-band changes during VIMS, the most consistent direction is an increase in low-frequency power. Of the 30 studies reporting a delta-band change, 16 (53.3%) report an increase, 6 a mixed pattern, 5 no change, and 3 a decrease. Of the 35 studies reporting a theta-band change, 20 (57.1%) report an increase, 6 a decrease, 5 a mixed pattern, and 4 no change. Increases in delta (1–4 Hz) and theta (4–8 Hz) power have been associated with the onset and increasing severity of cybersickness across diverse VR stimuli and hardware [33], [34], [43], [67], [71].

Alpha-band findings diverge across the corpus. Of the 39 studies reporting an alpha change, 15 (38.5%) report an increase, 11 (28.2%) report a decrease, 9 (23.1%) report a mixed pattern, and 4 (10.3%) report no change. Studies that report alpha suppression, particularly over central, parietal, and occipital cortices, interpret it as a marker of increased cognitive load under sensory conflict or as a consequence of postural-control demand [27], [53], [70]. Studies that report alpha enhancement attribute it to sensory gating or to drowsiness during prolonged exposure. subsection IV-F shows that the direction of the alpha change tracks VR display hardware, and reframes part of this corpus-level heterogeneity as hardware-attributable.

Higher-frequency bands show less consistent patterns than delta and theta. Of the 36 studies reporting a beta-band change, 11 (30.6%) report an increase, 10 (27.8%) a decrease, 10 a mixed pattern, and 5 no change. Individual reports range from beta suppression [71] to beta enhancement [109], and one study identifies beta as the most informative band for channel selection in classification [123]. Of the 21 studies reporting a gamma-band change, 7 (33.3%) report an increase, 6 no change, 5 a mixed pattern, and 3 a decrease. The full distribution of rhythm changes across anatomical regions is

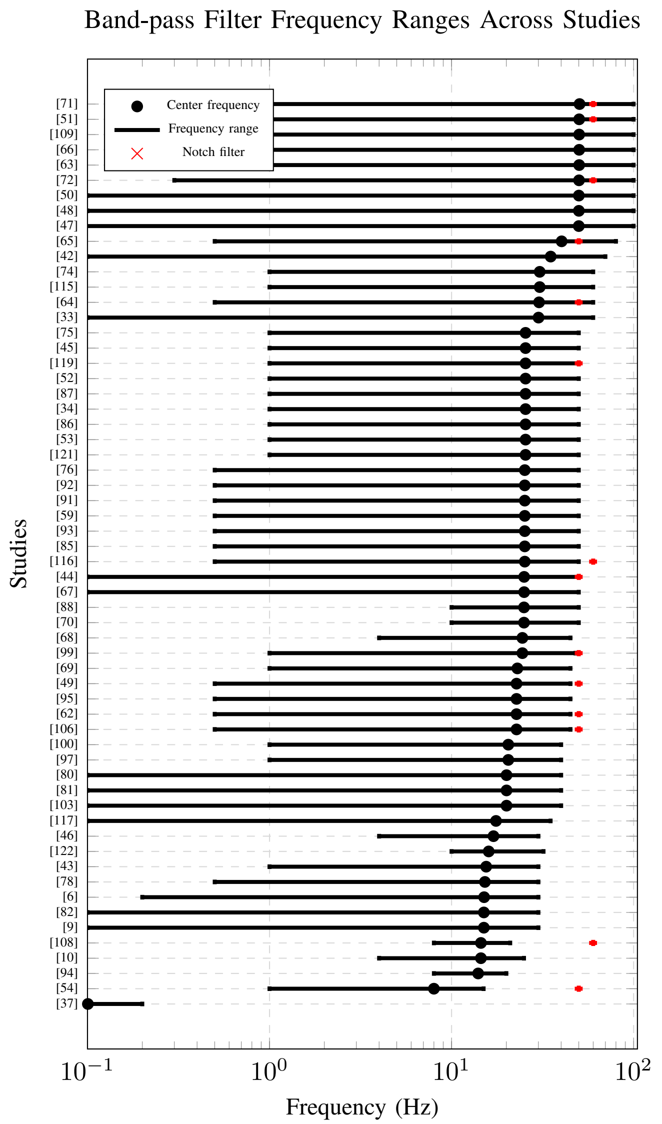


Fig. 6: Band-pass filter frequency ranges across studies showing center frequency (black dots) with frequency ranges (black error bars) and notch filter frequencies (red dots). The frequency range shows the low and high cutoff frequencies of the band-pass filters, while red crosses indicate notch filter frequencies used to remove specific interference (e.g., 50/60 Hz power line noise).

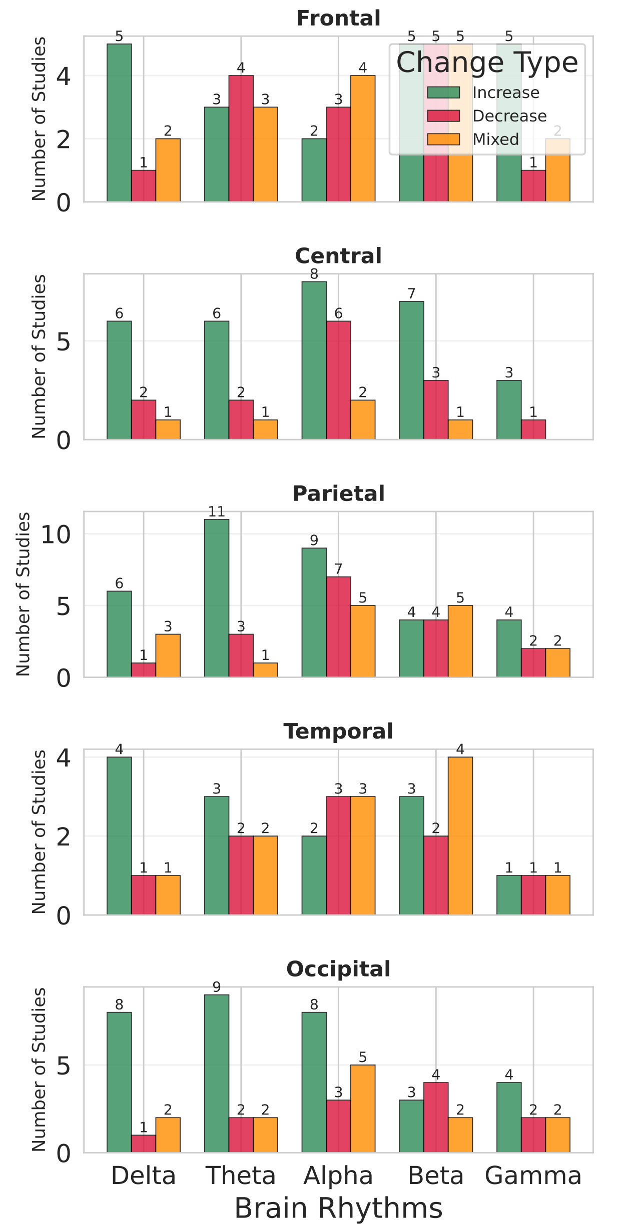


Fig. 7: Brain rhythm changes across anatomical regions during VIMS.

shown in Figure 7; as with alpha, the beta result is partly attributable to display hardware (subsection IV-F).

Test-retest reliability for delta, theta, and alpha power changes in frontal and central areas has been reported in one corpus study [85], and delta-band normalisation has been used as a marker of the post-exposure recovery time course [116]; the recovery literature is discussed in section V.

Beyond band-power, a subset of studies has examined additional spectral and information-theoretic features (entropy, complexity measures, amplitude modulation, mean theta-band frequency) [59], [89], [90], [104], [124]. Functional connectivity is investigated in 16 studies with mixed direction (7 mixed, 4 decrease, 3 increase, 2 no change), including transfer-entropy decreases at VIMS onset [33] and reduced connectivity between visual area V5 and vestibular regions [97]; phase-locking-value features have also been used as inputs to deep classification models [66], [106]. The causal relevance of these patterns is supported by intervention studies in which symptom mitigation correlates with normalisation of the same EEG bands [121], [125].

2) Event-Related Potentials and Evoked Responses: Beyond ongoing oscillations, a subset of studies has used event-related potentials (ERPs) to characterise how VIMS affects discrete cognitive processes. These studies indicate that VIMS is accompanied by measurable changes in attentional allocation and response inhibition. VIMS severity is associated with a reduction in the amplitude of the P3b component in a dual-task paradigm, a marker of diminished attentional resources [6]. VIMS has also been shown to impair response inhibition, as reflected by a larger N2 amplitude and a smaller, delayed P3 component during an oddball task [117]. Read alongside the sensory-conflict and cognitive-load accounts introduced in section II, these findings are consistent with VIMS consuming processing resources that would otherwise support task performance.

Susceptibility-related ERP patterns recur in studies that compare resistant and susceptible participants. VIMS-susceptible participants exhibit excessive parietal N2 amplitudes and impaired theta-band phase synchronisation compared to resistant counterparts [126], and individuals with high sickness levels show higher P3 amplitudes during sensory-mismatch conditions [42]. A separate ERP approach has linked the subjective experience of disorientation to a suppression of fronto-central heartbeat-evoked potentials, providing a link between interoceptive processing and a key symptom of cybersickness [51]. These susceptibility markers are revisited in subsection IV-D alongside the resting-state-prediction literature.

3) Hemodynamic Responses (fNIRS): Four fNIRS studies in the corpus report convergent regional hemodynamics during VIMS, providing a small but coherent body of evidence on the cortical regions involved. Bilateral angular-gyrus deactivation has been reported in association with subjective sickness scores [37], and increases in total hemoglobin concentration have been linked directly to VIMS onset [11], [38]. Reduced cortical activation during low-resolution stereoscopic exposure has also been documented [39].

The direction of the hemodynamic effect is not uniform across the small fNIRS literature: regional perfusion increases in some cortical areas coexist with decreases in others. Yamamura et al. also demonstrate the only fNIRS-based closed-loop mitigation system in the corpus, which is taken up in subsection IV-E.

C. Classification and Prediction of VIMS

Of the 92 corpus studies, 38 have classification or prediction of VIMS as their primary objective; a broader 41 use classification or regression methods as part of mixed-objective work. Corpus-level claims in this subsection are restricted to the 38-study primary-objective subset. Across these 38 studies, accuracy is the dominant reported performance metric, but sensitivity/specificity, AUC/F1, and the operational definition of the sick/non-sick label are reported in fewer than half of the subset.

1) Methodological Approaches and Feature Engineering: Computational approaches across the 38-study subset divide between traditional machine learning (15 of 38, 39.5%), deep learning (14, 36.8%), and regression (6, 15.8%); thresholding (1) and other approaches (2) account for the remainder. Traditional pipelines rely on hand-crafted features (Power Spectral Density, wavelet-derived energy ratios [78], spectral entropy and skew [59], time-domain statistics such as signal variance [94]) fed to Support Vector Machines, k-Nearest Neighbours, or Linear Discriminant Analysis [49], [127]; dimensionality reduction is addressed through Correlation Feature Selection [123] or genetic algorithms [128].

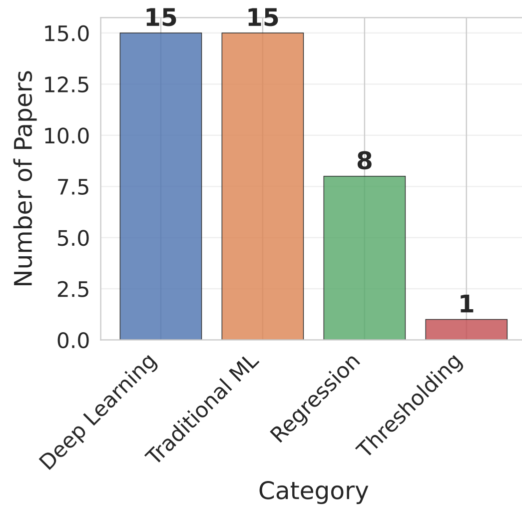


Fig. 8: Distribution of classification method categories across VIMS studies.

Deep-learning pipelines learn features directly from the data. Convolutional Neural Networks extract spatial and spectral features from EEG time series, spectrograms, or topographic scalp maps [68], with recurrent layers (Long Short-Term Memory [84], Gated Recurrent Units [93]) capturing temporal dynamics. Other architectures include spiking neural networks [120], shallow CNNs designed for real-time detection in mitigation systems [10], hybrid CNN–LSTM models [106], dual-pathway designs that process raw EEG and functional-connectivity inputs in parallel [66], and channel-attention mechanisms [63], [64]. Multimodal fusion combines EEG with peripheral physiological signals [105], with electromyography from postural muscles (reported as more informative than concurrent EEG in one study) [103], or with VR video features via transfer learning [72]. One paper investigates explainable-AI interpretation [105].

2) Reported Performance and Discrimination Metrics: Model validation predominantly employs k-fold cross-validation (21 of 38, 55.3%), with train-test splits (6), leave-one-subject-out cross-validation (1), and other or unspecified schemes (10) making up the remainder. The distribution of methods and validation schemes is summarised in Figure 8 and Figure 9.

Reported within-subject classification accuracies are high but rest on a small denominator: across the 8 of 38 studies that report a within-subject accuracy, the mean is 90.7% $\pm$ 8.4% (median 91.5%, range 73.3–100.0%). Cross-subject accuracies are reported by 22 of 38 studies, with a mean of 88.1% $\pm$ 11.6% (median 88.9%, range 55.0–99.7%). The two distributions are compared in Figure 10, and the temporal trend of reported accuracies is shown in Figure 11.

These headline accuracies are reported on heterogeneous discrimination metrics. Of the 38 classification studies, 13 (34.2%) report a headline accuracy with no sensitivity, specificity, AUC, or F1; only 7 (18.4%) report sensitivity and

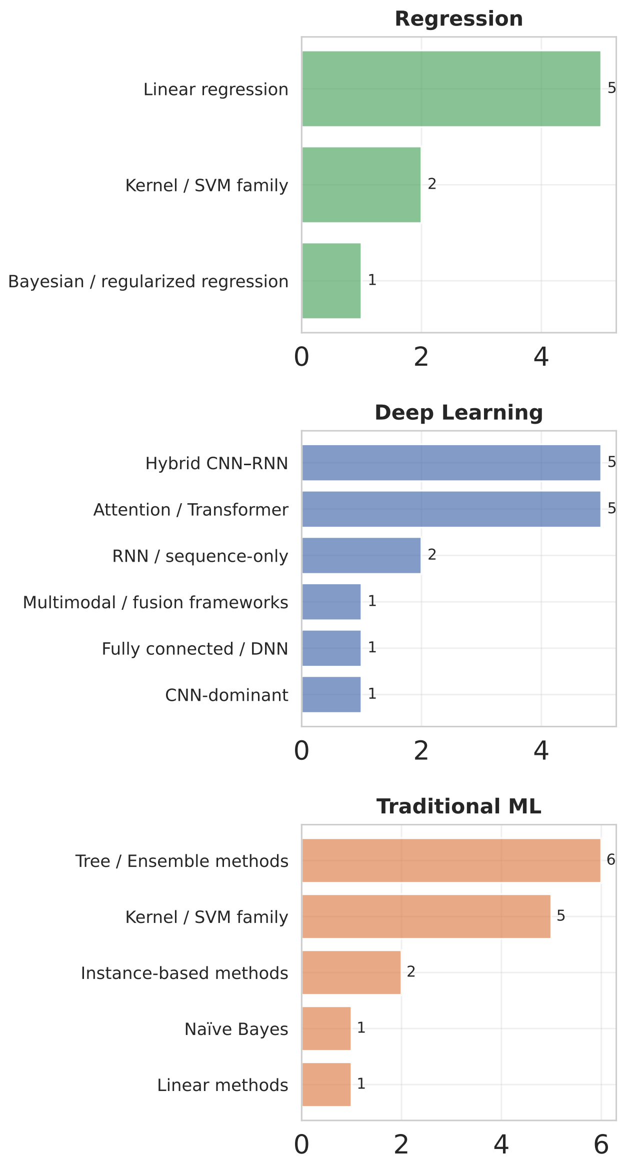


Fig. 9: Detailed breakdown of specific methods within each classification category.

specificity, and 10 (26.3%) report AUC or F1. Ten further studies (26.3%) report only regression metrics (R^2, RMSE) without a classification accuracy, and 2 (5.3%) report no quantitative performance.

3) Classification Tasks: Across the 38-study subset, binary classification of a "sick" versus "not sick" state is the most common task. Several studies report within-subject accuracies above 95% on this binary task [57], [63], [64], [68], [106], [127]. The sick/non-sick label that grounds these accuracies varies in how it is operationalised: 19 of 38 classification studies expose the threshold or its basis, of which 8 use a numeric cutoff and the remainder use self-reported binary labels, pre/post differences, or total-questionnaire-score cutoffs. The other 19 studies do not document the threshold at all.

A second task is multi-level classification, which categorises VIMS into several discrete levels of severity. Three-level classification has been demonstrated with multimodal fusion [9], [77] and with EEG alone using rule-based induction [100] or a genetic algorithm combined with a Support Vector Machine [87].

A third task is continuous prediction or regression of a VIMS severity score, measured by correlation coefficients and error metrics. Deep-learning models have demonstrated high correlations between predicted and actual SSQ scores [62], [65], [76], [91], [92], [119], while earlier work used Support Vector Regression [74], [115] and linear regression [129]. Addressing the threshold-heterogeneity problem raised above, one recent study develops continuous, calibration-free detection directly from physiological data with dynamic real-time ground-truth labels [130].

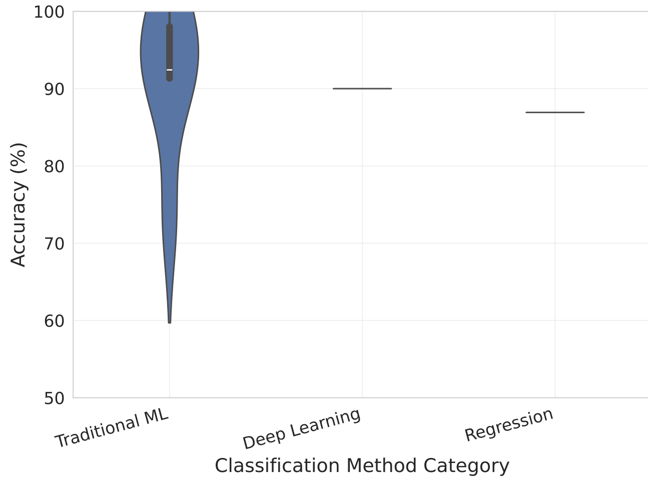
D. Individual Differences and Susceptibility

Resting-state neural features have been used to predict VIMS susceptibility before exposure, and during exposure susceptible and resistant individuals show distinct neural signatures. Demographic moderators are reported in a small subset of the corpus and warrant cautious interpretation given the small effect sizes documented in the broader literature.

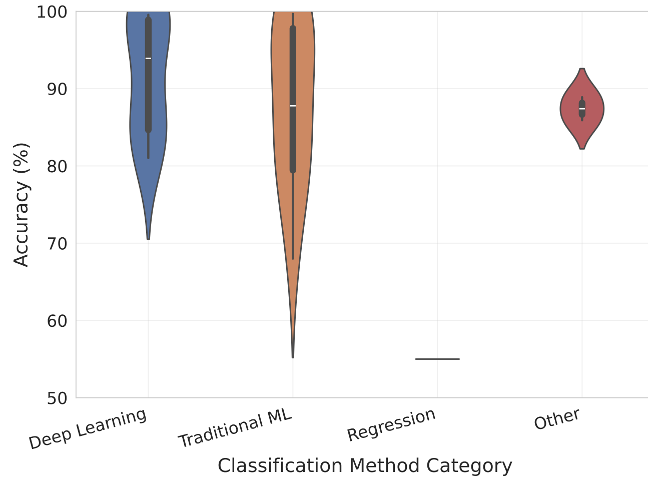
Pre-exposure prediction of VIMS has been demonstrated in three corpus studies. A spiking neural network classified subsequent sickness with 85.9% accuracy from pre-exposure resting-state EEG alone [120]. Resting-state EEG beta power in the left parietal region has been reported as a significant predictor of subsequent motion-sickness ratings in an in-car VR platform [82], and aperiodic components of the EEG signal have been reported as predictive of susceptibility to rotational motion [111]. These findings point to pre-existing neural traits that influence susceptibility, although the evidence base remains small.

During VIMS exposure, resistant and susceptible individuals show distinct neural signatures. Resistant individuals exhibit enhanced theta activity in the left parietal cortex, a candidate biomarker of resilience [81]. Susceptible individuals show excessive parietal N2 ERP amplitudes and impaired theta-band phase synchronisation [126], and higher P3 amplitudes during sensory-mismatch conditions [42]. One study reports that the cybersickness condition produces increased delta and decreased alpha power in both susceptible and non-susceptible groups, but susceptible participants additionally show lower absolute power in the beta and gamma bands at baseline [67]. An earlier study using a long-duration driving task found that the ratio of theta to total EEG power in frontal and parietal lobes differed between sick and non-sick groups, proposing this ratio as a physiological index of sensitivity [131].

Sex and gender differences are reported in 3 of 92 corpus studies, all from a single laboratory using a VR-neurofeedback paradigm [46]–[48]. Across all three studies, women reported greater cybersickness than men; in two of the three, higher cybersickness was additionally associated with poorer sensorimotor-rhythm neurofeedback performance [47], [48]. An additional 11 corpus studies use male-only samples (precluding sex comparisons) and 15 do not report participant sex composition, leaving 63 mixed-sex studies that report composition without analysing sex effects.



(a) Within-subject accuracy



(b) Cross-subject accuracy

Fig. 10: Classification accuracy distributions by method category for within-subject and cross-subject validation approaches.

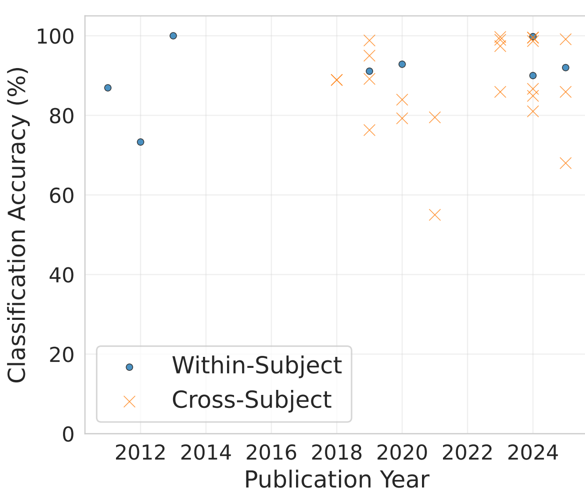


Fig. 11: Classification accuracy over time, comparing within-subject and cross-subject validation approaches.

Individual variability in VIMS susceptibility has measurable neural correlates and small demographic moderators. At present, the corpus is underpowered for either to be operationalised as a predictive biomarker. The mitigation literature reviewed next acts on the neural signals identified here.

E. Mitigation and Intervention Strategies

Mitigation research in the corpus splits into two threads: closed-loop adaptive VR that modifies the virtual environment in response to detected sickness (2 corpus studies), and interventions that target the user's brain state directly (5 corpus studies). Both threads are early-stage, and the small literature already documents one methodological obstacle specific to the closed-loop setting.

1) Closed-Loop Adaptive VR: Only two corpus studies implement closed-loop adaptation: the literature is in early-stage development. One system uses a two-stage shallow convolutional network to detect cybersickness from EEG and infer a likely causal factor, then selectively modifies the implicated VR parameter (navigation speed or scene complexity), producing a significant reduction in user-reported sickness compared to a control condition [10]. A second system uses fNIRS to monitor total hemoglobin concentration; on detecting increases correlated with sickness, the system dynamically reduces the user's field of view, demonstrating the feasibility of an fNIRS-based closed-loop pipeline [11].

2) Interventions Modulating Brain State: Five corpus studies investigate interventions intended to act on the underlying physiological state of the user, with concurrent EEG used to verify neural engagement. Transcutaneous auricular vagus nerve stimulation significantly reduced motion-sickness symptoms in one study, with the reduction correlated with stimulation-induced neural activation in specific brain regions measured by EEG source localisation [132]. Vibro-motor reprocessing therapy was associated with significant reductions in alpha, delta, and theta spectral power [125]. Providing synchronised auditory and physical motion cues during a simulated bicycle ride reduced VIMS scores relative to conditions with mismatched or absent cues, with the effect reflected in concurrent EEG recordings [121].

Two further studies, both from the same laboratory, investigated neurofeedback protocols in which users were trained to volitionally modulate their own brain activity. The presence of cybersickness symptoms was found to directly impair the user's ability to successfully perform the neurofeedback training task [48], and a placebo transcranial direct current stimulation procedure, while having no significant effect on reported sickness symptoms, paradoxically degraded performance on a concurrent neurofeedback task [47]. These findings illustrate a methodological challenge for any neuromarker-driven closed-loop intervention: the physiological state being targeted can interfere with the user's ability to engage respond to the intervention itself.

F. Corpus-Level Methodological Observations

Meta-analytic comparison across the 92 corpus studies indicates that VR display hardware is associated with EEG band-power responses and that stimulus-exposure duration is confounded with hardware. The findings in this subsection are observational; the interpretive critique and reporting recommendations they motivate are developed in section V.

Alpha-band VIMS responses differ between head-mounted-display and screen-based studies. The Mann–Whitney U test asks whether the ordered alpha-change codes differ between the two display classes; Cliff's δ gives the direction and size of that separation. This comparison yields Cliff's $\delta = 0.875$ ($p = 0.004$), indicating that head-mounted-display studies are more likely than screen-based studies to report alpha increases. Among head-mounted-display studies with a reported directional change, alpha increases occur four times as often as decreases (12 of 24 versus 3 of 24); screen-based studies in the same comparison report decreases only. This pattern reframes the alpha-band heterogeneity reported in subsection IV-B as partly hardware-attributable, rather than as a pure intersubject or task effect. The same head-mounted-display versus screen-based comparison for beta yields Cliff's $\delta = 0.664$ ($p = 0.015$), indicating the same direction of association beyond alpha. No corresponding between-device differences are detected for delta, theta, or gamma.

Stimulus-exposure duration is itself confounded with hardware. The Mann–Whitney comparison for exposure time shows that HMD studies use shorter exposure periods (mean 12.9 ± 14.8 minutes, $n = 62$) than non-HMD studies (mean 23.6 ± 15.1 minutes, $n = 21$; $U = 324.0$, $p < 0.001$, Cliff's $\delta = -0.502$, large effect). The hardware confound therefore covaries with how long the user is exposed, not only with the display optics; comparisons of band-power changes between HMD and screen studies necessarily mix the two effects.

These corpus-level observations, together with the inline reporting-gap counts in subsection IV-C and the baseline-heterogeneity findings in subsection IV-A, seed the design and reporting recommendations developed in section V.

V. Discussion

The prior scoping review of cybersickness EEG research by Chang et al. [8] reported delta-band increases, beta-band decreases, and inconsistent alpha-band findings; the present corpus refines this picture in two respects. The delta- and theta-band increases reproduce in the current corpus (16 of 30 studies for delta; 20 of 35 for theta). The reported alpha-band inconsistency partly resolves once VR display hardware is accounted for: head-mounted-display and screen-based studies differ with large effects on alpha (Cliff's $\delta = 0.875$, $p = 0.004$) and beta ($\delta = 0.664$, $p = 0.015$).

A. Objective Markers and the Limits of Self-Report

VIMS has no diagnostic biomarker, its ground truth is self-report. It is defined by its symptom profile, nausea, disorientation, and oculomotor disturbance, which are private experiences that cannot be externally verified [7], [17]; any candidate objective marker must therefore be calibrated against self-report rather than replacing it. Two corpus studies document functional cognitive consequences of this state: working-memory decrements scaling with reported severity [6] and inhibitory-control demand reflected in N2/P3 components [117], confirming that the state extends beyond subjective discomfort and carries measurable functional costs. The EEG and fNIRS correlates established in subsection IV-B are objective physiological measurements, but each was validated against self-reported severity and therefore presupposes the measure it would replace. Among the candidate objective markers, postural sway has a direct theoretical anchor in the postural-instability account, which predicts that pre-symptomatic sway covaries with subsequent severity [25], [26]; VR-specific reports document moderate coupling [9], [29], [133]. Pupillometry is a second candidate [134], [135], subject to arousal and cognitive-load confounds. Sixteen of the 92 corpus studies already co-record force-plate, balance-board, eye-tracker, or pupillometry alongside EEG; treating these signals as candidate outcome variables rather than covariates is a prerequisite for validating them as objective markers.

B. Methodological Challenges and Critical Perspective

Four classes of methodological issue emerge from the corpus and constrain cross-study synthesis in VIMS neuroimaging and neurostimulation: (1) classification reporting practices, (2) baseline selection, (3) protocol and hardware heterogeneity, and (4) conceptual and terminological variability.

Across the 38 classification or prediction studies in the corpus, three reporting practices limit the interpretability of headline performance numbers. Thirteen studies report a headline accuracy without sensitivity, specificity, AUC, or F1; under class imbalance, accuracy collapses toward the majority-class baseline and provides little information about the classifier's behaviour at decision-relevant operating points [136], [137]. Nineteen studies expose no sick-versus-non-sick threshold (neither a numeric cutoff nor a stated basis), and only eight give an explicit numeric cutoff converting SSQ or FMS scores into a label; the target variable is therefore operator-defined and not comparable across studies. One study reports leave-one-subject-out validation; twenty-one use k-fold cross-validation without subject stratification, which permits subject-identity leakage from training into the test fold and inflates apparent generalisation [138]. These three gaps are consistent with the high headline accuracies recurring in the corpus, including binary accuracies above 95% [57], [63], [64], [66], [106] and pre-post resting-state contrasts approaching 99% [49]: where accuracy is the only reported metric, the threshold defining the label is undocumented, and the validation scheme does not respect subject identity, the headline number does not support a generalisation claim. A related ground-truth issue compounds these reporting gaps: the SSQ is retrospective and coarse, and continuous motor in-task ratings such as joystick ratings risk encoding the motor act of reporting rather than the sickness state itself, so the very signal that supervises classification is heterogeneous before any modelling choice is made.

A second design issue concerns the baseline against which in-exposure neural change is measured. Closing the eyes elicits a large posterior alpha increase relative to eyes-open [139]–[141], so a study that contrasts VR exposure against an eyes-closed pre-task recording confounds the alpha-change estimate with a state shift unrelated to the visual stimulus. The VR exposure condition is, by construction, eyes-open inside a head-mounted display viewing a structured scene; an eyes-closed pre-task block is therefore not matched to the exposure condition on visual input, oculomotor state, or vigilance. This baseline choice is a minority practice in the corpus (5 of 77 EEG studies, or 6 if eyes-open + eyes-closed composites are counted), but distorts the alpha-band estimates of the affected studies, which typically present those estimates as primary outcomes.

Baseline choice is one of several protocol parameters that vary widely across the corpus, and the broader heterogeneity carries methodological consequences. At the corpus level, VR display hardware substantially modulates the band-power response: head-mounted-display and screen-based studies differ on the direction of alpha change with a large effect, and the alpha-band inconsistency reported in prior reviews [8] partly resolves once hardware is accounted for (see subsection IV-F). Stimulus exposure duration is itself confounded with display: head-mounted-display studies use shorter exposures than non-head-mounted-display studies so comparisons of band-power change between display classes necessarily mix the two effects. Recording quality varies similarly, ranging from research-grade systems to low-cost consumer-grade devices with fewer channels and lower signal quality [90], and the corpus is biased toward low-density montages: among the 72 EEG studies that disclose channel count, 58 (80.6%) use 32 or fewer channels and 14 (19.4%) use 10 or fewer. Low-density montages are appropriate for applied paradigms that target well-characterised localised signals; for exploratory characterisation of VIMS neurophysiology they have insufficient spatial resolution for reliable source localisation. Reporting quality compounds the hardware-protocol heterogeneity: of the 77 EEG corpus studies, 43 (55.8%) do not state the reference electrode and 59 (76.6%) do not state a notch-filter frequency, which limits both reproducibility and meta-analytic re-pooling. Hardware, stimulus, montage, and reporting parameters vary study by study; requiring several to match at once leaves little overlap. In the corpus, the largest subgroup that aligns on sickness measure (SSQ), display (head-mounted display), stimulus class (game), and modality (EEG) contains only five of the 92 corpus studies. Cross-study generalisation is therefore difficult: the field has no replicable majority paradigm, and statements that pool results across the literature aggregate category labels rather than a shared experimental design. Samples are small and demographically narrow (typically young university students), and 11 of the 92 studies recruited male-only samples [67], [125], which limits statistical power for individual-difference questions including the sex/gender effects discussed in subsection V-C. Finally, the experiences and stimuli are themselves heterogeneous: passive 360-degree video [109] is not equivalent to active interactive navigation [27], and findings from head-mounted-display, screen, CAVE, and smartphone-VR systems [85], [125], [142] are often discussed interchangeably.

Finally, the field exhibits conceptual variability in both the use of theory and the use of terminology. Most corpus studies (73 of 92 explicitly, 14 implicitly) cite an etiological theory of VIMS in their introduction or methods, but few return to that framework when interpreting their neurophysiological findings; a neural feature observed in a given paradigm could reflect sensory conflict, postural control, expectation-related vection, display optics, or some combination, and without an explicit theoretical commitment the literature cannot adjudicate between these possibilities. Terminology compounds the interpretive ambiguity: motion sickness, visually induced motion sickness, simulator sickness, and cybersickness are related but non-identical constructs tied to different exposure contexts (see section II), yet they are frequently used interchangeably [14], [15], [17], [18]. A further consequence of these practices is the labelling of any statistically significant feature as a "neuromarker" without establishing its specificity, sensitivity, or reliability across contexts.

Taken together, these methodological and conceptual issues constrain cross-study synthesis and reproducibility.

C. Future Directions

The methodological challenges documented in section V translate into six concrete directions for future work: standardised reporting and open data, causal mechanisms, generalisation and ecological validation, sex- and gender-stratified design, objective behavioural markers, and longitudinal characterisation of adaptation and recovery.

First, standardised reporting represents a gap documented across the corpus. For classification and prediction studies, the minimum items follow from the gaps documented in section V: a discriminative metric (AUC or balanced accuracy) with 95% confidence interval, sensitivity and specificity at a stated operating point, the numeric label threshold, a subject-respecting validation scheme, and comparison against a trivial and a non-neural feature baseline. For studies that quantify in-exposure neural change, a baseline matched to the exposure condition on visual input, oculomotor state, and vigilance – with the choice and its rationale reported – is required for reproducibility; the available options are discussed in section V. For EEG, the reference electrode and notch-filter frequency are required for reproducibility and meta-analytic re-pooling. For subjective sickness measurement, reporting the questionnaire instrument, its administration time-point relative to exposure end, and the numeric cutoff used to assign class labels enables cross-study comparability; where continuous in-task ratings are used, the response modality and any associated motor artefact exclusion strategy are necessary for reproducibility. For stimulus and display, the device model, field-of-view, and frame rate alongside the stimulus class (passive or active, recorded or interactive); adoption of a small number of canonical reference paradigms, at minimum a passive visual-flow exposure and an active first-person navigation task, would allow direct cross-laboratory comparison without the confound of idiosyncratic stimulus design.

These reporting standards are a precondition for, but not a substitute for, shared infrastructure: community-curated benchmark datasets with multimodal recordings (EEG, EOG, ECG, head motion, sway, eye tracking), well-defined subjective labels, and coverage of both passive and active paradigms. A small number of corpus papers have already released public datasets [56], [72], [130]; releasing analysis code and pre-trained models alongside the data would further support replication and cumulative analysis.

Second, the causal status of documented neural associations remains largely unexamined. The literature has documented associations between specific neural patterns and VIMS across multiple studies, but whether these associations are causal has not been established. Non-invasive brain stimulation offers a direct test: transcranial alternating-current stimulation can target specific oscillations (e.g., alpha or theta), and existing corpus studies that combine stimulation with VR exposure [47], [110] provide a methodological template for asking whether modulating a candidate neuromarker changes sickness severity. Coupled with computational models of multisensory integration and predictive coding, interventional designs can begin to distinguish neural features that drive the VIMS experience from features that merely accompany it.

Third, generalisation and ecological validation remain open challenges for models built for VIMS. In the four corpus studies that report both within- and cross-subject accuracy, cross-subject performance is lower in each case, and only one of 38 classification studies reports a leave-one-subject-out evaluation; transfer-learning and domain-adaptation approaches that explicitly model inter-subject and inter-session variability address a current gap if brain-directed methods are to operate without per-user calibration. Validation beyond the laboratory is a further open direction: at-home, naturalistic paradigms such as the in-home gaming setup of [143] address the question of whether laboratory neuromarkers survive the noise of consumer-grade hardware and uncontrolled environments.

Fourth, sex and gender represent an underexplored factor in the corpus. Of the 92 corpus studies, 11 used male-only samples, 15 did not report sex composition, and only three (all from a single laboratory) reported higher cybersickness in women [46]–[48]. The broader literature on sex differences in VIMS susceptibility is mixed: [144] argues that reported effects are modest in magnitude and plausibly mediated by interpupillary-distance fit, prior gaming experience, and questionnaire framing; [135] reports that male–female differences disappear when prior gaming experience is included as a covariate; and [145] caution that the claim of greater female susceptibility is overstated in the simulation literature. Taken together, the three corpus reports and the broader literature are consistent with a modest, variable effect whose magnitude and moderators cannot be resolved at individual-study scale, and whose apparent direction may partly reflect gaming-experience differences and questionnaire framing rather than a stable sex difference in susceptibility. Reporting sex-stratified outcomes (band-power changes, connectivity, classification metrics), even when single-study samples are underpowered for inferential testing, would enable subsequent meta-analytic re-pooling. Logging interpupillary-distance setting and gaming experience as covariates would address both the sex confound and the hardware effects documented in subsection IV-F. Adequately powered designs – pooled multi-site cohorts or longitudinal within-subject designs – are the precondition for resolving the magnitude and moderators of a sex effect on VIMS.

Fifth, objective behavioural markers represent an underdeveloped line of investigation. The candidate markers introduced in section V are already collected by 16 of the 92 corpus studies, but typically as covariates rather than as candidate ground truth. Postural sway is the strongest theoretically anchored candidate: reporting a coupling estimate between pre-symptomatic sway dynamics and subsequent self-reported sickness, alongside the protocol metadata that this estimate depends on (stance, surface, sampling rate), would anchor the construct across laboratories. Pupillometry can supplement self-report subject to arousal and cognitive-load confounds, and requires controls on ambient luminance and on task demand. Heart-rate variability, electrodermal activity, and head-motion dynamics are further candidates whose corpus-level predictive value is not yet quantified; multi-modal integration that fuses behavioural, autonomic, and neural signals offers a more holistic state estimate than any single modality. Treating these markers as outcome variables, rather than as covariates, is the precondition for shifting VIMS research toward objective ground truth.

Finally, longitudinal designs represent a gap in the current corpus. The corpus is dominated by single-session experiments, leaving the neurodynamics by which users adapt to VR (“getting one’s VR legs”) largely uncharacterised, and the single corpus recovery study [116] is insufficient to support strong claims about residual neural state. Multi-session within-subject designs would enable tracking of how the EEG signatures identified in this review evolve with repeated exposure, and mapping the dissipation of post-exposure signatures onto behaviour.

VI. Conclusion

This systematic review synthesized 92 studies on the neurophysiology of visually induced motion sickness in VR, uniting EEG-based correlate and classification work with fNIRS hemodynamic evidence and non-invasive neurostimulation interventions under a common neuromarker frame.

The corpus converges on a low-frequency EEG signature of VIMS (increased delta and theta power) that reproduces across diverse stimuli, hardware, and laboratories, and is consistent with the pattern reported by the prior scoping review of Chang et al. [8]. Alpha-band findings have been inconsistent in the literature; the meta-analytic comparison reported here attributes a substantial portion of that inconsistency to VR display hardware, with head-mounted-display and screen-based studies systematically differing in the direction of alpha and beta responses. This hardware confound, compounded by differences in stimulus-exposure duration, limits the extent to which band-power results can be pooled across the literature.

The corpus also documents attentional and inhibitory-control costs during VIMS in event-related potential studies, regional hemodynamic correlates in fNIRS, and pre-exposure resting-state predictors of individual susceptibility.

Early closed-loop systems (adaptive VR pipelines and direct brain-state interventions) demonstrate that neurophysiological detection of VIMS is sufficient in principle to drive real-time mitigation, though the evidence base for both threads remains small.

Alongside these findings, the review identifies methodological barriers that constrain generalisation: sparse and heterogeneous reporting in the classification literature, baseline-selection practices that confound neural estimates, hardware and protocol heterogeneity that prevent direct cross-study replication, and terminological variability that complicates theoretical interpretation. The causal status of the observed neural associations, cross-subject and ecological generalisation of detection models, the role of sex and gender, and the neurodynamics of adaptation and recovery over repeated exposure remain open questions in the current literature.

Acknowledgments

This research was funded by the European Union ERANET CHIST-ERA 2020 (CHIST-ERA-20-BCI-003 - "GENESIS") and the French Research Agency (ANR-21-CHRA-0001-01).

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